

>TYPE E-Z1.ASM
EXEMPLARY BSTAM INTERFACE AS GUIDE TO GENERALIZATION

THIS VERSION WILL WORK WITH ZSIO (PORT B)

VERSION ZSIO/1.0 JUNE 10,1981

```
BEGIN      ORG      103H
INITIL     JMP      INITIU      ;UART/USART INITIATION ENTRY POINT
INSPORT    JMP      INSPRU      ;STATUS PORT READ ENTRY POINT
ERRSET     JMP      ERRSTU      ;UART/USART ERROR RESET ENTRY POINT
INPORT     JMP      INPRTU      ;READ DATA PORT ENTRY POINT
OUTPORT    JMP      OUTPRU      ;WRITE DATA PORT ENTRY POINT
DUART      JMP      DUARTU
DUARTU     RET
```

IF YOU WISH TO USE OTHER REGISTERS SUCH AS HL,BC,DE
(NOT INCLUDING SP REGISTER)
BE SURE TO PUSH BSTAM'S REGISTERS ON TO STACK AND POP
THEM BACK OFF BEFORE RETURNING TO BSTAM.
THERE IS AMPLE ROOM TO PUSH ALL REGISTERS ONTO BSTAM'S
STACK, PLUS ROOM FOR 10 LEVELS OF CALL'S.

THERE ARE 300 BYTES OF USER SPACE AVAILABLE TO YOU.

THE JUMP'S CODED AT THE BEGINNING OF THIS PROGRAM
CANNOT BE MOVED IN ANY WAY.

GOOD LUCK AND WELCOME TO THE REAL WORLD OF
TELECOMMUNICATIONS.

```
TPS      EQU      17H      ;STATUS PORT (CHANNEL B)
TP        EQU      15H      ;DATA PORT
```

```
INITIU    EQU      $
```

THIS IS THE UART/USART INITIATION ROUTINE.
TO BE COMPATIBLE WITH MOST BSTAM USERS USE FOLLOWING
INITIATION GUIDE LINES:

1. USE 1 STOP BIT (OPTIONAL - 2)
2. USE 8 DATA BITS (MUST)
3. USE 1 START BIT (MUST)
4. USE 16X FOR CLOCK RATE (MUST)
5. USE ASYNCHRONOUS MODE ONLY (MUST)

```
CALL      SIGNON      ;TELL USER WHAT'S IMPLEMENTED

MVI       A,18H        ;CHANNEL RESET
OUT       TPS
```



```

MVI    A,4          ;PTR TO WR4
OUT     TPS
MVI    A,44H        ;X16 - 1 STOP BIT - NO PARITY
OUT     TPS
MVI    A,3          ;PTR TO WR3
OUT     TPS
MVI    A,0C1H       ;RX 8 DATA - RX ENABLE
OUT     TPS
MVI    A,5          ;PTR TO WR5
OUT     TPS
MVI    A,0EAH       ;DTR - TX 8 DATA - TX ENABLE - RTS
OUT     TPS
MVI    A,11H        ;RESET INT - PTR TO WR1
OUT     TPS
XRA     A           ;ALL INT. ARE DISABLED
OUT     TPS

```

```

MVI    A,0CH        ; SET DTR ON COMM BD HIGH
OUT     06CH        ; .SEND IT TO CMD REG

MVI    A,041H       ; TELL SYSTEM THAT BAUD RATE IS INTERNAL
OUT     06CH        ; SELECT INTERNAL

MVI    A,06EH       ; TX 9600 BAUD
OUT     06CH        ; CMD PORT

MVI    A,07EH       ; RX 9600 BAUD
OUT     06CH        ; COMM CMD PORT

RET                     ;RETURN TO BSTAM

```

NSPRU EQU \$

THIS IS THE STATUS READ PORT ROUTINE.
WHEN EXITING THIS ROUTINE BSTAM EXPECTS IN REGISTER A
THE FOLLOWING BITS TO BE SET IF NEEDED:

1. 20 BIT SET IF FRAMING ERROR
2. 10 BIT SET IF OVERRUN ERROR
3. 08 BIT SET IF PARITY ERROR
4. 04 BIT SET IF TRANSMITTER EMPTY

IF YOUR UART/USART DOES NOT HAVE TRANSMITTER
EMPTY FEATURE, THEN ALWAYS SET THIS BIT ON.

5. 02 BIT SET IF RECEIVER READY
6. 01 BIT SET IF TRANSMITTER READY
7. DO NOT SET THE 80 BIT OR 40 BIT

```

PUSH    B
IN       TPS
MOV      C,A
MVI      B,4          ;PRE-SET TRANSMITTER EMPTY BIT
ANI      1            ;CHK DATA RECEIVED
JZ       CHKTXRD
MVI      A,2          ;SET RECEIVER READY BIT
ORA      B
MOV      B,A
CHKTXRD EQU $
MOV      A,C
ANI      4            ;CHK TRANSMITTER READY
JZ       DONE
MVI      A,1          ;SET TRANSMITTER READY
ORA      B

```

```

MOV      B,A
DONE     EQU      $
MOV      A,B
POP      B
RET      ;RETURN TO BSTAM
;BITS ARE NOW SET FOR BSTAM

```

```

ERRSTU   EQU      $

```

```

THIS IS THE ERROR RESET FOR UART/USART ROUTINE.
YOUR UART/USART MAY NOT NEED TO BE RESET IF FRAMING,
OVERRUN, OR PARITY ERROR IS FOUND. IF THIS IS YOUR
CASE, JUST DO A RETURN.

```

```

RET      ;RETURN TO BSTAM

```

```

INPRTU   EQU      $

```

```

THIS IS THE READ DATA PORT ROUTINE.
BEFORE THIS ROUTINE IS ENTERED THE 02 BIT OF
STATUS READ ROUTINE MUST HAVE BEEN SET.
DO NOT CLEAR THE 80 BIT FROM THE DATA INPUT PORT.
RETURN WITH REGISTER A LOADED WITH INPUT DATA.

```

```

IN      TP      ;GET DATA FROM PORT
RET      ;RETURN TO BSTAM

```


OUTPRU EQU \$

THIS IS THE WRITE DATA PORT ROUTINE.
BEFORE THIS ROUTINE IS ENTERED THE 04 BIT AND 01 BIT
OF STATUS READ MUST BE SET.
DO NOT CLEAR THE 80 BIT FROM THE DATA OUTPUT PORT.
REGISTER A CONTAINS THE OUTPUT DATA.

OUT TP ;SEND DATA TO REMOTE CPU
RET ;RETURN TO BSTAM

MESSGE: DB 0DH,0AH,0DH,0AH
DB 'VERSION LANIER EZ-1/1.0 JUNE 10, 1981',0DH,0AH
DB 'ZSIO ON PORT B USING NO PARITY',0DH,0AH
DB 'USING PORT 015H AS DATA,017H AS STATUS',0AH,0DH,'\$'

SIGNON: PUSH B
PUSH D
PUSH H
LXI D,MESSGE
MVI C,9
CALL 5 ;ANOUNCE VERSION TO USER
POP H
POP D
POP B
RET

END BEGIN

```
B>TYPE UZSIO.ASM
;EXEMPLARY BSTAM INTERFACE AS GUIDE TO GENERALIZATION
;
;THIS VERSION WILL WORK WITH ZSIO (PORT B)
;
;VERSION ZSIO/1.0 JUNE 10,1981
;
BEGIN      ORG        103H
INITIL     JMP        INITIU                ;UART/USART INITIATION ENTRY POINT
INSPORT    JMP        INSPRU               ;STATUS PORT READ ENTRY POINT
ERRSET     JMP        ERRSTU              ;UART/USART ERROR RESET ENTRY POINT
INPORT     JMP        INPRTU              ;READ DATA PORT ENTRY POINT
OUTPORT    JMP        OUTPRU              ;WRITE DATA PORT ENTRY POINT
DUART      JMP        DUARTU
DUARTU     RET

;
;
;
;          IF YOU WISH TO USE OTHER REGISTERS SUCH AS HL,BC,DE
;          (NOT INCLUDING SP REGISTER)
;          BE SURE TO PUSH BSTAM'S REGISTERS ON TO STACK AND POP
;          THEM BACK OFF BEFORE RETURNING TO BSTAM.
;          THERE IS AMPLE ROOM TO PUSH ALL REGISTERS ONTO BSTAM'S
;          STACK, PLUS ROOM FOR 10 LEVELS OF CALL'S.
;
;          THERE ARE 300 BYTES OF USER SPACE AVAILABLE TO YOU.
;
;          THE JUMP'S CODED AT THE BEGINNING OF THIS PROGRAM
;          CANNOT BE MOVED IN ANY WAY.
;
;          GOOD LUCK AND WELCOME TO THE REAL WORLD OF
;          TELECOMMUNICATIONS.
;
;
;
TPS         EQU        1FH                ;STATUS PORT (CHANNEL B)
TP          EQU        1EH                ;DATA PORT
;
;
;
;
INITIU     EQU        $

;          THIS IS THE UART/USART INITIATION ROUTINE.
;          TO BE COMPATIBLE WITH MOST BSTAM USERS USE FOLLOWING
;          INITIATION GUIDE LINES:
;          1. USE 1 STOP BIT           (OPTIONAL - 2)
;          2. USE 8 DATA BITS         (MUST)
;          3. USE 1 START BIT          (MUST)
;          4. USE 16X FOR CLOCK RATE   (MUST)
;          5. USE ASYNCHRONOUS MODE ONLY (MUST)
;
CALL        SIGNON                        ;TELL USER WHAT'S IMPLEMENTED
MVI        A,18H                          ;CHANNEL RESET
OUT        TPS
MVI        A,4                            ;PTR TO WR4
```



```

OUT      TPS
MVI      A,44H      ;X16 - 1 STOP BIT - NO PARITY
OUT      TPS
MVI      A,3        ;PTR TO WR3
OUT      TPS
MVI      A,0C1H     ;RX 8 DATA - RX ENABLE
OUT      TPS
MVI      A,5        ;PTR TO WR5
OUT      TPS
MVI      A,0EAH     ;DTR - TX 8 DATA - TX ENABLE - RTS
OUT      TPS
MVI      A,11H      ;RESET INT - PTR TO WR1
OUT      TPS
XRA      A          ;ALL INT. ARE DISABLED
OUT      TPS
RET                      ;RETURN TO BSTAM

```

```

INSPRU EQU $
THIS IS THE STATUS READ PORT ROUTINE.
WHEN EXITING THIS ROUTINE BSTAM EXPECTS IN REGISTER A
THE FOLLOWING BITS TO BE SET IF NEEDED:
1. 20 BIT SET IF FRAMING ERROR
2. 10 BIT SET IF OVERRUN ERROR
3. 08 BIT SET IF PARITY ERROR
4. 04 BIT SET IF TRANSMITTER EMPTY
   IF YOUR UART/USART DOES NOT HAVE TRANSMITTER
   EMPTY FEATURE, THEN ALWAYS SET THIS BIT ON.
5. 02 BIT SET IF RECEIVER READY
6. 01 BIT SET IF TRANSMITTER READY
7. DO NOT SET THE 80 BIT OR 40 BIT

```

```

PUSH     B
IN       TPS
MOV      C,A
MVI      B,4        ;PRE-SET TRANSMITTER EMPTY BIT
ANI      1          ;CHK DATA RECEIVED
JZ       CHKTXRD
MVI      A,2        ;SET RECEIVER READY BIT
ORA      B
MOV      B,A
CHKTXRD EQU $
MOV      A,C
ANI      4          ;CHK TRANSMITTER READY
JZ       DONE
MVI      A,1        ;SET TRANSMITTER READY
ORA      B
MOV      B,A
DONE EQU $
MOV      A,B        ;BITS ARE NOW SET FOR BSTAM
POP      B
RET                      ;RETURN TO BSTAM

```

```

ERRSTU EQU $

```

THIS IS THE ERROR RESET FOR UART/USART ROUTINE.
YOUR UART/USART MAY NOT NEED TO BE RESET IF FRAMING,
OVERRUN, OR PARITY ERROR IS FOUND. IF THIS IS YOUR
CASE, JUST DO A RETURN.

RET ;RETURN TO BSTAM

INPRTU EQU \$

THIS IS THE READ DATA PORT ROUTINE.
BEFORE THIS ROUTINE IS ENTERED THE 02 BIT OF
STATUS READ ROUTINE MUST HAVE BEEN SET.
DO NOT CLEAR THE 80 BIT FROM THE DATA INPUT PORT.
RETURN WITH REGISTER A LOADED WITH INPUT DATA.

IN TP ;GET DATA FROM PORT
RET ;RETURN TO BSTAM

OUTPRU EQU \$

THIS IS THE WRITE DATA PORT ROUTINE.
BEFORE THIS ROUTINE IS ENTERED THE 04 BIT AND 01 BIT
OF STATUS READ MUST BE SET.
DO NOT CLEAR THE 80 BIT FROM THE DATA OUTPUT PORT.
REGISTER A CONTAINS THE OUTPUT DATA.

OUT TP ;SEND DATA TO REMOTE CPU
RET ;RETURN TO BSTAM

MESSAGE: DB ODH,0AH,ODH,0AH
DB 'VERSION ZSIO/1.0 JUNE 10, 1981',ODH,0AH

DB 'ZSIO ON PORT B USING NO PARITY',ODH,DAH,'\$'

SIGNON: PUSH B

PUSH D

PUSH H

LXI D,MESSGE

MVI C,9

CALL 5 ;ANOUNCE VERSION TO USER

POP H

POP D

POP B

RET

END BEGIN

TITLE 'BSTAM/BSTMS INTERFACE FOR THE ARCHIVES BUSINESS COMPUTER'

THIS VERSION WILL OPERATE WITH AN ARCHIVES BUSINESS COMPUTER
USING CHANNEL 'A' OF THE ZILOG SIO CHIP.

VERSION: UARC/SIO JUNE 5, 1980
WRITTEN BY: JAMES SIEM --- ARCHIVES INCORPORATED
404 W. 35TH. STREET
DAVENPORT, IOWA 52806

MACLIB Z80 ;LIBRARY OF Z-80 SUPPORT MACROS

BEGIN ORG 103H ;START OF THE UART/USART CUSTOMIZATION MODULE
INITIL JMP INITIU ;UART/USART INITIATION ROUTINE ENTRY POINT
INSPORT JMP INSPRU ;UART/USART STATUS ROUTINE ENTRY POINT
ERRSET JMP ERRSTU ;UART/USART ERROR RESET ROUTINE ENTRY POINT
INPORT JMP INPRTU ;READ DATA ROUTINE ENTRY POINT
OUTPORT JMP OUTPRU ;WRITE DATA ROUTINE ENTRY POINT
UART JMP DUARTU
DUARTU RET

PS EQU 0EDH ;SIO STATUS PORT
P EQU 0ECH ;SIO DATA PORT
RESET EQU 030H ;SIO ERROR RESET COMMAND CODE

PAGE

INITIU EQU \$
THIS IS THE UART/USART INITIATION ROUTINE.
TO BE COMPATIBLE WITH MOST BSTAM USERS I USED THE
FOLLOWING INITIALIZATION GUIDE LINES:
1. 1 STOP BIT
2. 8 DATA BITS
3. 1 START BIT
4. 16X CLOCK RATE
5. ASYNCHRONOUS MODE
6. NO PARITY

TITLE 'BSTAM/BSTMS INTERFACE FOR THE ARCHIVES BUSINESS COMPUTER'

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INPORT JMP INPRU ;READ DATA ROUTINE ENTRY POINT
OUTPORT JMP OUTPRU ;WRITE DATA ROUTINE ENTRY POINT
DUART JMP DUARTU
DUARTU RET

TPS EQU OEDH ;SIO STATUS PORT
TP EQU DECH ;SIO DATA PORT
ERESSET EQU O30H ;SIO ERROR RESET COMMAND CODE

PAGE

INITIU EQU \$
THIS IS THE UART/USART INITIATION ROUTINE.
TO BE COMPATIBLE WITH MOST BSTAM USERS I USED THE
FOLLOWING INITIALIZATION GUIDE LINES:
1. 1 STOP BIT
2. 8 DATA BITS
3. 1 START BIT
4. 16X CLOCK RATE
5. ASYNCHRONOUS MODE
6. NO PARITY
#


```

PUSH    B                ;SAVE <BC>
PUSH    D                ;SAVE <DE>
PUSH    H                ;SAVE <HL>
LXI     D,MESSEGE        ;GET THE SIGN-ON MESSAGE ADDRESS
MVI     C,009H           ;LOAD THE CP/M PRINT MESSAGE FUNCTION CODE
CALL    00005H           ;ANNOUNCE THE VERSION TO THE USER
MVI     C,TPS            ;GET THE PORT ADDRESS TO INITIALIZE
LXI     H,TABLE          ;GET THE ADDRESS OF THE BYTE TABLE
MVI     B,TLEN           ;GET THE LENGTH OF THE BYTE TABLE
OTIR                    ;AND SEND OUT THE BLOCK OF DATA
POP     H                ;RECOVER <HL>
POP     D                ;RECOVER <DE>
POP     B                ;RECOVER <BC>
RET                                ;DONE

```

```

TABLE:  DB    018H                ;SIO INITIALIZATION BYTE TABLE
        DB    004H
        DB    044H
        DB    003H
        DB    0E1H
        DB    005H
        DB    0EAH

```

```

TLEN    EQU    $-TABLE        ;COMPUTE THE TABLE LENGTH

```

PAGE

```

INSPRU  EQU    $
;
; THIS IS THE UART/USART STATUS ROUTINE.
; WHEN EXITING THIS ROUTINE BSTAM RECEIVES
; IN REGISTER 'A' THE FOLLOWING:
; 1. 20 BIT SET IF FRAMING ERROR
; 2. 10 BIT SET IF OVERRUN ERROR
; 3. 08 BIT SET IF PARITY ERROR
; 4. 04 BIT SET IF TRANSMITTER EMPTY
; 5. 02 BIT SET IF RECEIVER READY
; 6. 01 BIT SET IF TRANSMITTER READY
; 7. 80 BIT AND 40 BIT ALWAYS RESET (ZERO)
;

```

```

PUSH    B                ;SAVE <BC>
MVI     B,000H           ;ZAP <B>
IN      TPS              ;GET THE FIRST SIO STATUS WORD
BIT     0,A              ;TEST THE CHARACTER AVAILABLE BIT
JRZ     INSPRU1          ;JUMP IFF NO CHARACTER IS AVAILABLE NOW

```

```

        SETB     1,B                ;ELSE SET THE 02 BIT IN REG <B>
INSPRU1:BIT     2,A                ;THEN TEST THE TRANSMITTER EMPTY BIT
        JRZ     INSPRU2            ;JUMP IFF TRANSMITTER IS FULL
        SETB     0,B                ;ELSE SET THE TRANSMITTER READY BIT
        SETB     2,B                ;AND SET THE TRANSMITTER EMPTY BIT
INSPRU2:MVI     A,001H            ;LOAD THE SIO REGISTER NUMBER TO READ NEXT
        OUT     TPS                ;AND SELECT IT
        IN      TPS                ;THEN GET THE SECOND SIO STATUS WORD
        RAR     ;ROTATE IT INTO THE REQUIRED BIT POSITIONS
        ANI     038H              ;MASK OFF THE IRRELEVANT BITS
        ORA     B                  ;MERGE IT INTO THE PREVIOUS STATUS WORD RESULTS
        POP     B                  ;RECOVER <BC>
        RET     ;DONE

```

PAGE

```

ERRSTU EQU     $                  ;THIS IS THE UART/USART ERROR RESET ROUTINE.

        MVI     A,ERESET          ;GET THE SIO ERROR RESET CODE
        OUT     TPS                ;SEND IT OUT TO THE SIO CHIP
        RET     ;DONE

```

```

INPRTU EQU     $                  ;THIS IS THE DATA INPUT ROUTINE.

        IN      TP                ;GET THE CURRENT DATA FROM THE SIO
        RET     ;DONE

```

```

OUTPRU EQU     $                  ;THIS IS THE DATA OUTPUT ROUTINE.

        OUT     TP                ;SEND THE CURRENT DATA TO THE SIO
        RET     ;DONE

```

```

MESSAGE: EQU     $                  ;THIS IS THE CURRENT SIGN-ON MESSAGE

        DB      0DH,0AH,0DH,0AH

```


DB 'VERSION ARC/SIO 4.4 JUNE 5,1980',ODH,DAH
DB 'SIO ON PORT ECD/EDS USING NO PARITY',ODH,DAH,'\$'

END BEGIN

^L

Programming

The system program first issues a series of commands that initialize the basic mode of operation and then other commands that qualify conditions within the selected mode. For example, the asynchronous mode, character length, clock rate, number of stop bits, even or odd parity might be set first; then the interrupt mode; and finally, receiver or transmitter enable.

Both channels contain registers that must be programmed via the system program prior to operation. The channel-select input (B/ $\bar{A}$) and the control/data input (C/ $\bar{D}$) are the command-structure addressing controls, and are normally controlled by the CPU address bus. Figures 15 and 16 illustrate the timing relationships for programming the write registers and transferring data and status.

Read Registers. The SIO contains three read registers for Channel B and two read registers for Channel A (RR0-RR2 in Figure 13) that can be read to obtain the status information; RR2 contains the internally-modifiable interrupt vector and is only in the Channel B register set. The status information includes error conditions, interrupt vector and standard communications-interface signals.

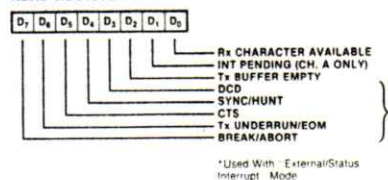
To read the contents of a selected read register other than RR0, the system program must first write the pointer byte to WR0 in exactly the same way as a write register operation. Then, by executing a read instruction, the contents of the addressed read register can be read by the CPU.

The status bits of RR0 and RR1 are carefully grouped to simplify status monitoring. For example, when the interrupt vector indicates that a Special Receive Condition interrupt has occurred, all the appropriate error bits can be read from a single register (RR1).

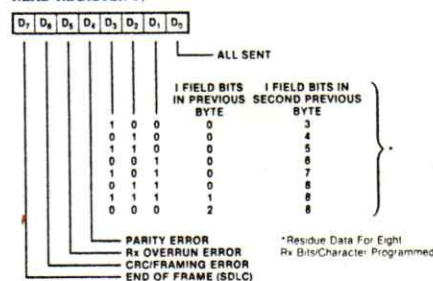
Write Registers. The SIO contains eight write registers for Channel B and seven write registers for Channel A (WR0-WR7 in Figure 14) that are programmed separately to configure the functional personality of the channels; WR2 contains the interrupt vector for both channels and is only in the Channel B register set. With the exception of WR0, programming the write registers requires two bytes. The first byte is to WR0 and contains three bits (D_0 - D_2) that point to the selected register; the second byte is the actual control word that is written into the register to configure the SIO.

WR0 is a special case in that all of the basic commands can be written to it with a single byte. Reset (internal or external) initializes the pointer bits D_0 - D_2 to point to WR0. This implies that a channel reset must not be combined with the pointing to any register.

READ REGISTER 0



READ REGISTER 1†



READ REGISTER 2*

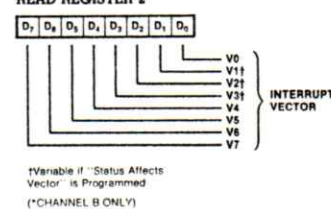


Figure 13. Read Register Bit Functions

Programming (Continued)

WRITE REGISTER 0

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	REGISTER 0
0	0	0	0	0	0	0	1	REGISTER 1
0	0	0	0	0	0	1	0	REGISTER 2
0	0	0	0	0	0	1	1	REGISTER 3
0	0	0	0	0	1	0	0	REGISTER 4
0	0	0	0	0	1	0	1	REGISTER 5
0	0	0	0	0	1	1	0	REGISTER 6
0	0	0	0	0	1	1	1	REGISTER 7
0	0	0	0	0	0	0	0	NULL CODE
0	0	0	0	0	0	0	1	SEND ABORT (SDLC)
0	0	0	0	0	0	1	0	RESET EXT/STATUS INTERRUPTS
0	0	0	0	0	0	1	1	CHANNEL RESET
0	0	0	0	0	1	0	0	ENABLE INT ON NEXT R _x CHARACTER
0	0	0	0	0	1	0	1	RESET T _x INT PENDING
0	0	0	0	0	1	1	0	ERROR RESET
0	0	0	0	0	1	1	1	RETURN FROM INT (CH A ONLY)
0	0	0	0	0	0	0	0	NULL CODE
0	0	0	0	0	0	0	1	RESET R _x CRC CHECKER
0	0	0	0	0	0	1	0	RESET T _x CRC GENERATOR
0	0	0	0	0	0	1	1	RESET T _x UNDERRUN/EOM LATCH

WRITE REGISTER 1

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	EXT INT ENABLE
0	0	0	0	0	0	0	1	T _x INT ENABLE
0	0	0	0	0	0	1	0	STATUS AFFECTS VECTOR (CH. B ONLY)
0	0	0	0	0	0	1	1	R _x INT DISABLE
0	0	0	0	0	1	0	0	R _x INT ON FIRST CHARACTER
0	0	0	0	0	1	0	1	INT ON ALL R _x CHARACTERS (PARITY AFFECTS VECTOR)
0	0	0	0	0	1	1	0	INT ON ALL R _x CHARACTERS (PARITY DOES NOT AFFECT VECTOR)
0	0	0	0	0	0	0	0	WAIT/READY ON R/T
0	0	0	0	0	0	0	1	WAIT/READY FUNCTION
0	0	0	0	0	0	1	0	WAIT/READY ENABLE

*On On Special Condition

WRITE REGISTER 2 (CHANNEL B ONLY)

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	V0
0	0	0	0	0	0	0	1	V1
0	0	0	0	0	0	1	0	V2
0	0	0	0	0	0	1	1	V3
0	0	0	0	0	1	0	0	V4
0	0	0	0	0	1	0	1	V5
0	0	0	0	0	1	1	0	V6
0	0	0	0	0	1	1	1	V7

INTERRUPT VECTOR

WRITE REGISTER 3

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	R _x ENABLE
0	0	0	0	0	0	0	1	SYNC CHARACTER LOAD INHIBIT
0	0	0	0	0	0	1	0	ADDRESS SEARCH MODE (SDLC)
0	0	0	0	0	0	1	1	R _x CRC ENABLE
0	0	0	0	0	1	0	0	ENTER HUNT PHASE
0	0	0	0	0	1	0	1	AUTO ENABLES
0	0	0	0	0	0	0	0	R _x 5 BITS/CHARACTER
0	0	0	0	0	0	0	1	R _x 7 BITS/CHARACTER
0	0	0	0	0	0	1	0	R _x 6 BITS/CHARACTER
0	0	0	0	0	0	1	1	R _x 8 BITS/CHARACTER

WRITE REGISTER 4

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	PARITY ENABLE
0	0	0	0	0	0	0	1	PARITY EVEN/ODD
0	0	0	0	0	0	0	0	SYNC MODES ENABLE
0	0	0	0	0	0	0	1	1 STOP BIT/CHARACTER
0	0	0	0	0	0	1	0	11% STOP BITS/CHARACTER
0	0	0	0	0	0	1	1	2 STOP BITS/CHARACTER
0	0	0	0	0	0	0	0	8 BIT SYNC CHARACTER
0	0	0	0	0	0	0	1	16 BIT SYNC CHARACTER
0	0	0	0	0	0	1	0	SDLC MODE (01111110 FLAG)
0	0	0	0	0	0	1	1	EXTERNAL SYNC MODE
0	0	0	0	0	0	0	0	X1 CLOCK MODE
0	0	0	0	0	0	0	1	X16 CLOCK MODE
0	0	0	0	0	0	1	0	X32 CLOCK MODE
0	0	0	0	0	0	1	1	X64 CLOCK MODE

WRITE REGISTER 5

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	T _x CRC ENABLE
0	0	0	0	0	0	0	1	RTS
0	0	0	0	0	0	1	0	SDLC/CRC-16
0	0	0	0	0	0	1	1	T _x ENABLE
0	0	0	0	0	0	0	0	SEND BREAK
0	0	0	0	0	0	0	0	T _x 5 BITS (OR LESS)/CHARACTER
0	0	0	0	0	0	0	1	T _x 7 BITS/CHARACTER
0	0	0	0	0	0	1	0	T _x 6 BITS/CHARACTER
0	0	0	0	0	0	1	1	T _x 8 BITS/CHARACTER

DTR

WRITE REGISTER 6

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	SYNC BIT 0
0	0	0	0	0	0	0	1	SYNC BIT 1
0	0	0	0	0	0	1	0	SYNC BIT 2
0	0	0	0	0	0	1	1	SYNC BIT 3
0	0	0	0	0	1	0	0	SYNC BIT 4
0	0	0	0	0	1	0	1	SYNC BIT 5
0	0	0	0	0	1	1	0	SYNC BIT 6
0	0	0	0	0	1	1	1	SYNC BIT 7

*Also SDLC Address Field

WRITE REGISTER 7

D₇ D₆ D₅ D₄ D₃ D₂ D₁ D₀

0	0	0	0	0	0	0	0	SYNC BIT 8
0	0	0	0	0	0	0	1	SYNC BIT 9
0	0	0	0	0	0	1	0	SYNC BIT 10
0	0	0	0	0	0	1	1	SYNC BIT 11
0	0	0	0	0	1	0	0	SYNC BIT 12
0	0	0	0	0	1	0	1	SYNC BIT 13
0	0	0	0	0	1	1	0	SYNC BIT 14
0	0	0	0	0	1	1	1	SYNC BIT 15

*For SDLC It Must Be Programmed to 01111110 For Flag Recognition

Figure 14. Write Register Bit Functions

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